Diego Tyner

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EDUCATION

University of California, Davis | Davis, CA — March 2026
B.S. Computer Science (High Honors) and Cognitive Science (Honors), Minor in Neuroscience — **GPA: 3.95**
Relevant Courses: Artificial Intelligence, Deep Learning, Machine Learning, Algorithm Design & Analysis

- Citation for Outstanding Performance in Cognitive Science

SKILLS

Languages: Python, JavaScript/TypeScript, HTML/CSS, Java, Kotlin, C, C++, C#, Bash
AI / LLM: Gemini API, transformers, pgvector, RAG pipelines, spaCy, NLTK
Frameworks: FastAPI, Flask, Node.js, Express.js, React, Next.js, Vite, Tailwind
ML / Data: TensorFlow, PyTorch, scikit-learn, NumPy, Pandas, OpenCV, Matplotlib
Databases: PostgreSQL (SQL), MongoDB (NoSQL), Supabase
Tools / Devops: Git/GitHub, Docker, Linux, Windows, Drupal, Codeium
Spoken Languages: English (native), Spanish (fluent - CA Seal of Biliteracy)

EXPERIENCE

Web Developer, *Alfalfa and Forage Research Team | Davis, CA* — Oct 2024 - Jun 2026

- Sole developer for two production research websites, serving 16,775+ users and 107,000+ tracked events since Feb 2025
- Developed and maintained a Drupal/MySQL site and an ASP.NET site using HTML/CSS, JavaScript, C#, and .NET
- Built crawler-based link-checking tooling to detect and fix broken links (404s) and legacy bugs from a 2021 domain migration, contributing to 48% growth across variety trial pages, the site's most-used content (H1 2026 vs. H2 2025)

PROJECTS

RAG News Dashboard, *Personal Project* | [Demo] · [Code] — Jul 2025 - Sep 2025

- Architected a dockerized FastAPI backend and Next.js frontend for a full-stack news analysis platform
- Implemented a RAG pipeline end-to-end, chunking and embedding articles into PostgreSQL/pgvector and retrieving via cosine similarity, and grounded Gemini LLM responses with context for hallucination-reduced query answering
- Enriched retrieval context with integrated news scraping/extraction pipeline, bias detection, and NLP-based summarization to support the RAG-driven query system

BCI Wheelchair Controller (Weimo), *Davis Neurotech* | [Poster] · [Code] — Nov 2025 - Jun 2026

- Multimodal brain–computer interface (BCI) wheelchair prototype fusing EEG motor imagery, LiDAR mapping, eye tracking, and motor actuation for hands-free navigation for users with neurodegenerative conditions
- Built the PyQt6-based control application coordinating 8 concurrent processes (EEG streaming/classification, LiDAR, pathfinding, head tracking, YOLOv8 detection, motor driver) via per-field-locked shared memory for low-latency access
- Implemented real-time head/eye tracking and EEG preprocessing/classification pipelines powering the live demo
- Won UC Davis Neurotech competition (10 teams) and was selected to present at the California Neurotech Conference

Canvas Resource Semantic Search, *Personal Project* | [Code] — Mar 2025 - Apr 2025

- Engineered an end-to-end pipeline to authenticate, scrape, and ingest Canvas course files and lecture transcripts using Selenium and concurrent downloaders
- Extracted and chunked text from course texts, generating vector embeddings (PyTorch, MiniLM) for semantic representation of course content
- Indexed embeddings in PostgreSQL/pgvector, allowing semantic retrieval across a student’s full course library

Real Finder, *Personal Project* | [Live Site**]** · [Code] — Sep 2025 - May 2026

- Engineered a repeatable OCR ingestion pipeline (PaddleOCR, PyMuPDF, rapidfuzz) to extract, clean, and fuzzy-match song entries against an external chord database, deduplicating aliased titles
- Delivered fuzzy search, in-app PDF viewing, and persistent personal setlists on the frontend